Space-Invariant Shape Descriptors for CAD Mesh Retrieval

What each descriptor does, how it is computed, and why it is needed

QRCAD: training-free CAD shape retrieval

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This document describes fifteen descriptors that turn an STL mesh into a vector which does not change when the part is moved, rotated or scaled. They are arranged as a *ladder*. For every descriptor there is a concrete pair of mechanical parts that **all of the descriptors before it fail to tell apart**, and the document explains in plain English what the new descriptor measures that makes the difference. Each descriptor has four parts: a plain-English explanation, the mathematics, an algorithm, and the "why it is needed" example with a descriptor-by-descriptor comparison. Where an example could be simulated, the separation was measured numerically on synthetic parts (§1.4).

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1 Introduction

1.1 The goal

A retrieval system has to turn each CAD part (an STL triangle mesh) into a fixed-length vector of numbers, its *descriptor*, so that similar parts get similar vectors and a nearest-neighbour search can find them. The same part must give the same vector however it happens to be placed in the file: shifted, rotated, or exported in millimetres instead of inches. No single measurement can do this for every kind of part. The descriptor is therefore built from several *blocks*, each measuring a different kind of geometric information. This document explains every block, in the order in which each one adds information the earlier blocks cannot capture.

1.2 Invariance: what "space-invariant" means

Let the surface be the point set $X \subset \mathbb{R}^3$. A move of the part is a similarity transform $g(\mathbf{p}) = s\mathbf{R}\mathbf{p} + \mathbf{t}$ with scale $s > 0$, rotation $\mathbf{R} \in \mathrm{SO}(3)$ and shift $\mathbf{t} \in \mathbb{R}^3$. A descriptor f is *space-invariant* when

$$f(gX) = f(X) \qquad \text{for every such } g. \tag{1}$$

In practice this holds automatically if the descriptor is built only from:

- **differences** of positions $\mathbf{p}_i - \mathbf{p}_j$, or positions measured from the part's own centre (which removes $\mathbf{t}$);
- **lengths, angles and dot products** of such vectors, and unit normals $\hat{\mathbf{n}}$ (which removes $\mathbf{R}$);
- **ratios**: every length is divided by a length of the part itself, every area by a length squared, and so on (which removes s).

1.3 Common ingredients

Uniform surface samples. Most blocks look at random points on the surface. A point is drawn uniformly by choosing triangle t with probability A_t / A (its share of the total area) and then placing it at

$$\mathbf{p} = (1 - \sqrt{r_1})\,\mathbf{x}_0 + \sqrt{r_1}(1 - r_2)\,\mathbf{x}_1 + \sqrt{r_1}\,r_2\,\mathbf{x}_2, \qquad r_1, r_2 \sim U[0,1], \tag{2}$$

where $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$ are the triangle's corners. Each sample also keeps the unit normal $\hat{\mathbf{n}}_t$ of its triangle. This makes the samples independent of how finely the CAD exporter tessellated the part.

Reference length. All lengths are divided by the *diameter* of the sample set,

$$D_\varnothing = \max_{i,j} \left\| \mathbf{p}_i - \mathbf{p}_j \right\|. \tag{3}$$

It is found from the convex hull of the samples. $D_\varnothing$ does not change under rotation or shift and scales exactly with the part, so every normalised distance lies in $[0, 1]$. Curvatures and eigenvalues are normalised with $\sqrt{A}$ instead, where A is the total surface area.

Histograms. Most blocks are histograms with B bins. Each is divided by its total so that it sums to 1, and then passed through the square root, $\mathbf{h} \mapsto \sqrt{\mathbf{h}}$ (the Hellinger map). With this map the inner product of two blocks equals the Bhattacharyya similarity of the two distributions, which is the natural way to compare them (§7.2).

Mesh connectivity. The edge, curvature, spectral and topological blocks need a welded mesh: coincident STL corners merged into shared vertices, and a map from each edge to the two faces that meet at it.

1.4 How the "why it is needed" examples are judged

Each descriptor section ends with a pair of parts, **Part A** and **Part B**, and a table listing what every earlier descriptor sees. Three kinds of evidence are used.

1. Identical by construction (same). Some pairs are built so that an earlier descriptor gives *exactly* the same vector. An example is two parts whose surface normals have exactly the same directions and areas.

2. Changes only in proportion to a small region (nearly same). Many pairs differ only in a small region of the surface. The following principle then bounds what an area-based descriptor can see.

Small-feature principle

Suppose Parts A and B are identical except on a region covering a fraction ε of the surface area, and the difference does not change the reference length. Take any descriptor built as a histogram of some function of k independent random surface points: $k = 1$ for D1, $k = 2$ for D2, $k = 3$ for triangles, $k = 4$ for D4. Then

$$\|F_A - F_B\|_1 \;\leq\; 2k\varepsilon. \tag{4}$$

Reason: draw the samples for both parts together so that each one coincides unless it falls in the differing region, which happens with probability ε. A k-tuple differs with probability at most $k\varepsilon$, and the L1 distance between two distributions is at most twice the probability that their samples differ.
The same bound holds for edge histograms weighted by edge length, with ε the fraction of edge length that differs. **In plain words:** a feature that covers 1% of the surface can move D2 by at most 4%, however important the feature is to an engineer.

3. Measured separation (SNR). Where the parts could be simulated exactly (analytic surfaces with exact normals), the sampling-based descriptors were computed numerically. The table reports

$$\mathrm{SNR} = \frac{\text{mean L1 distance between Part A and Part B}}{\text{mean L1 distance between two independent resamplings of the same part}},$$

using $n = 2 \times 10^4$ surface samples and $S = 2 \times 10^5$ random tuples per part, with three independent resamplings of each part. SNR ≈ 1 **means the descriptor cannot tell the parts apart** (the difference is no larger than resampling noise). An SNR of 10 or more is a clear separation.

1.5 The descriptor ladder at a glance

#	Descriptor	New kind of information it adds
1	D1 radial distance	how far the surface lies from the centre
2	D2 pair distance	how far apart pieces of the surface are from each other
3	Triangle (A3 + D3)	whether the surface is locally line-like or spreads out in two directions
4	D4 tetrahedron volume	whether the surface is flat or bulges into 3-D; handedness
5	SHPS orientation spectrum	which directions the surface faces
6	NCF normal correlation	how facing directions relate to distance (front vs. back of a wall)
7	LBE vibration spectrum	how the surface is connected over long paths (slits, cuts)
8	DAD edge angles	how sharp the folds between faces are
9	Curvature (SI + CV)	how smooth faces bend: convex or concave, cylinder or sphere, tight or gentle
10	CRS-R ring profile	diameters about the part's own axis; concentricity
11	T1 genus	how many through-holes the part has
12	PHD persistent homology	where the holes are and how large they are
13	CRS-A angular pattern	how features are spaced around the axis (tooth, bolt, spline patterns)
14	PAF planar families	how many parallel machined planes exist, coplanarity
15	MTS wall thickness	how thick the material is between two surfaces

2 Where the surface is: point-set statistics

The first four descriptors look only at *positions* of random surface points. They are cheap, very robust to bad meshes, and capture the overall size distribution and proportions of a part.

2.1 D1: radial distance distribution

vector length 32

In plain English

Stand at the centre of the part and measure the distance to every piece of its surface. D1 is the histogram of those distances, with the largest one scaled to 1. A ball gives a single spike, because every surface point is equally far from the centre. A long shaft gives a wide spread, from its middle (close) to its ends (far). A hollow ball shows two spikes, one for the outside and one for the inside surface.

Mathematics. With the area-weighted surface centroid $\mathbf{c} = \frac{1}{A}\sum_t A_t \mathbf{b}_t$, where $\mathbf{b}_t$ is the centre of triangle t,

$$d_i = \|\mathbf{p}_i - \mathbf{c}\|, \qquad \hat{d}_i = \frac{d_i}{\max_j d_j} \in [0,1], \qquad F_{\mathrm{D1}} = \mathrm{hist}_{32}\left(\{\hat{d}_i\}; [0,1]\right). \tag{5}$$

The centre moves with the part and the distances are unchanged by rotation. Dividing by the largest distance removes scale.

Reference shapes. Sphere: all mass at $\hat{d} = 1$. Cube: support $[1/\sqrt{3},\, 1] \approx [0.58, 1]$, from face centres to corners.

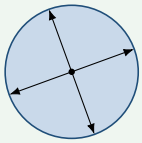
Algorithm 1 D1

Input: surface samples $\{\mathbf{p}_i\}_{i=1}^{n}$, triangle areas and centres

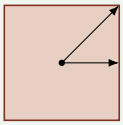
1: $\mathbf{c} \leftarrow \sum_t A_t \mathbf{b}_t / A$
2: $d_i \leftarrow \|\mathbf{p}_i - \mathbf{c}\|$ for all i; $\hat{d}_i \leftarrow d_i / \max_j d_j$
3: **return** L1-normalised 32-bin histogram of $\{\hat{d}_i\}$ on $[0,1]$

Why it is needed: the baseline

D1 is the first rung, so there is no earlier descriptor to beat. It is the cheapest possible summary of the overall form.



A: steel ball



B: gauge block (cube)

D1 separates them at once. The ball puts 100% of its mass in the last bin; the cube spreads its mass from 0.58 (face centres) to 1 (corners). D1 also separates a solid ball from a hollow one, because the inner wall appears as a second spike at r_{in}/R.

What it still cannot see

D1 records only *how far* each piece of surface is from the centre, not *in which direction*. Rotating one arm of a part about the centre leaves D1 exactly unchanged.

2.2 D2: pairwise distance distribution

vector length 32

In plain English

Pick two random points on the surface and measure the straight-line distance between them. Repeat this a few hundred thousand times and histogram the distances. A compact part gives mostly short distances. A long part gives many long ones. A part made of separated lumps gives one bump for every distance between lumps. D2 does not need a centre at all, only the part itself.

Mathematics.

$$\hat{d}_{ij} = \frac{\|\mathbf{p}_i - \mathbf{p}_j\|}{D_\varnothing} \in [0,1], \qquad F_{\text{D2}} = \text{hist}_{32}\left(\{\hat{d}_{ij}\}; [0,1]\right) \tag{6}$$

over S random pairs. Differences cancel the shift, norms are unchanged by rotation, and $D_\varnothing$ removes scale.

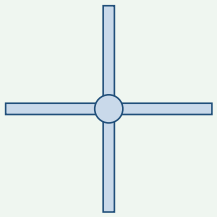
Reference shape. For a sphere of radius R the distance density is linear, $p(d) = d/(2R^2)$ on $[0, 2R]$. This follows from Archimedes' theorem: the cosine of the angle between two random points on a sphere is uniform on $[-1, 1]$.

Algorithm 2 D2

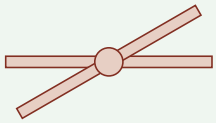
Input: samples $\{\mathbf{p}_i\}$, reference length $D_\varnothing$, number of pairs S

1: draw S index pairs (i, j), $i \neq j$
2: $\hat{d} \leftarrow \|\mathbf{p}_i - \mathbf{p}_j\| / D_\varnothing$
3: **return** L1-normalised 32-bin histogram on $[0, 1]$

Why it is needed: evenly spaced vs. paired spokes



A: evenly spaced (90°)



B: paired (0, 30, 180, 210°)

Both are 4-spoke spiders with identical spokes. In A the spokes are evenly spaced; in B they form two narrow V's.

Descriptor	Result on this pair, and why
D1	**same** (SNR 0.9). Both spiders are balanced, so the centre is the hub in both, and every spoke contributes exactly the same set of distances from the centre. D1 is identical by construction.
D2	**different** (SNR 18). Tip-to-tip distances differ. A has $4 \times 1.41L$ and $2 \times 2L$, while B has $2 \times 0.52L$, $2 \times 1.93L$ and $2 \times 2L$ (L = spoke length).

What it still cannot see

D2 records the distance between two points but not whether the surface is line-like or spreads out in two directions. Gently bowing a thin rod changes its distances only at second order (a 6 mm bow on a 200 mm rod changes lengths by about 0.24%).

2.3 Triangle descriptor: A3 (angle) and D3 (area)

vector length 32 + 32

In plain English

Pick three random surface points and look at the triangle they form. A3 records the angle at one corner and D3 records the triangle's area. Points on a straight rod always form needle-thin triangles, with angles near 0° or 180° and almost no area. As soon as the surface spreads out in a second direction, the triangles open up: angles fill the range and areas grow. A3 needs no reference length at all. D3 carries the size information that A3 lacks.

Mathematics. With $\mathbf{u} = \mathbf{p}_2 - \mathbf{p}_1$ and $\mathbf{w} = \mathbf{p}_3 - \mathbf{p}_1$,

$$\theta = \operatorname{atan2}\left(\|\mathbf{u} \times \mathbf{w}\|, \mathbf{u} \cdot \mathbf{w}\right) \in [0, \pi], \qquad F_{\mathrm{A3}} = \operatorname{hist}_{32}(\{\theta\}; [0, \pi]), \tag{7}$$

$$\hat{A} = \frac{\frac{1}{2}\|\mathbf{u} \times \mathbf{w}\|}{D_{\varnothing}^2} \in \left[0, \tfrac{\sqrt{3}}{4}\right], \qquad F_{\mathrm{D3}} = \operatorname{hist}_{32}\left(\{\sqrt{\hat{A}}\}; \left[0, \sqrt{\sqrt{3}/4}\right]\right). \tag{8}$$

The upper bound $\sqrt{3}/4$ is the area of an equilateral triangle with side $D_{\varnothing}$, the largest triangle possible inside the part. The square root turns area into a length-like quantity and spreads the histogram evenly. Using atan2 rather than arccos keeps angles near 0 and π accurate.

Algorithm 3 Triangle descriptor

Input: samples $\{\mathbf{p}_i\}$, $D_\varnothing$, number of triples S

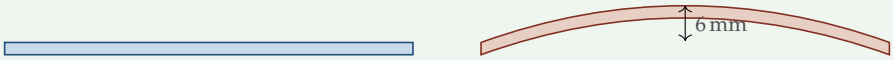
draw S index triples (i, j, k) with distinct entries

$\mathbf{u} \leftarrow \mathbf{p}_j - \mathbf{p}_i$; $\mathbf{w} \leftarrow \mathbf{p}_k - \mathbf{p}_i$; $\mathbf{x} \leftarrow \mathbf{u} \times \mathbf{w}$

$\theta \leftarrow \operatorname{atan2}(\|\mathbf{x}\|, \mathbf{u} \cdot \mathbf{w})$; $\quad a \leftarrow \sqrt{\frac{1}{2}\|\mathbf{x}\|}/D_\varnothing$

return $[\mathrm{hist}_{32}(\theta),\ \mathrm{hist}_{32}(a)]$, each L1-normalised

Why it is needed: straight vs. bowed rod



A: straight pin, $\varnothing 4 \times 200$ **B: same pin, bowed 6 mm**

The pair is a straight push-rod or dowel against the same rod with a 6 mm bow.

Descriptor	Result on this pair, and why
D1	**nearly same** (SNR 1.2). Distances from the centre change only at second order in the bow, $\approx 8s^2/(3L^2) = 0.24\%$.
D2	**nearly same** (SNR 1.0). Same second-order argument.
A3	**different** (SNR 11). On the straight rod almost every triangle is needle-like, with angles near 0° or 180°. The bow lifts the middle 6 mm out of line, so triangles open up.
D3	**different** (SNR 26). On the straight rod areas are capped by the rod radius (2 mm). The bow allows triangle heights up to 6 mm, so areas roughly triple.

The reason: bending changes *distances* only at second order, but it changes triangle *heights* at first order.

What it still cannot see

Triangles cannot tell whether the surface leaves a *plane*. A flat strip and a slightly cambered strip have almost the same triangle areas and angles, because the areas change only at second order.

2.4 D4: random tetrahedron volume (with handedness)

vector length 32 + 1

In plain English

Pick four random surface points and compute the volume of the tetrahedron they span. Four points on a flat face always give zero volume, so D4 is a direct test of "how three-dimensional" the part is: sheet-metal blanks sit near zero, while castings and bulky parts spread out. The *sign* of the volume also tells a left-handed part from its mirror image, as long as the four points are taken in a fixed order (nearest to farthest from the centre). No other descriptor in this document can do that.

Mathematics. With the four points ordered by increasing distance from $\mathbf{c}$,

$$V = \tfrac{1}{6}\det\left[\mathbf{p}_2 - \mathbf{p}_1 \mid \mathbf{p}_3 - \mathbf{p}_1 \mid \mathbf{p}_4 - \mathbf{p}_1\right], \qquad \hat{V} = \frac{|V|}{D_\varnothing^3} \in \left[0, \tfrac{1}{6\sqrt{2}}\right], \tag{9}$$

$$F_{\mathrm{D4}} = \left[\mathrm{hist}_{32}\left(\{\sqrt[3]{\hat{V}}\}; [0, (6\sqrt{2})^{-1/3}]\right),\ \gamma_1(\{V/\sigma_V\})\right]. \tag{10}$$

Here $1/(6\sqrt{2})$ is the volume of a regular tetrahedron with edge $D_\varnothing$, and the cube root again gives a length-like quantity. γ_1 is the skewness of the signed volumes. $\det(\mathbf{R}M) = \det M$ for rotations, so

V is rotation invariant, while a mirror flips its sign. For any part with a plane of mirror symmetry, $\gamma_1 = 0$. In a numerical test with three unequal orthogonal arms, $\gamma_1 = -1.28$ for the part and $+1.28$ for its mirror image. With *unordered* points, $\gamma_1 \approx 0$ for both, which is why the ordering matters.

Algorithm 4 D4

Input: samples $\{\mathbf{p}_i\}$, centre $\mathbf{c}$, $D_\varnothing$, number of quadruples S

1: draw S quadruples; sort each by $\|\mathbf{p} - \mathbf{c}\|$ (ascending)
2: $V \leftarrow \frac{1}{6}\left((\mathbf{p}_2 - \mathbf{p}_1) \times (\mathbf{p}_3 - \mathbf{p}_1)\right) \cdot (\mathbf{p}_4 - \mathbf{p}_1)$
3: **return** $\text{hist}_{32}(\sqrt[3]{|V|}/D_\varnothing)$ and the skewness of V

Why it is needed: flat vs. cambered strip



A: flat strip $200 \times 60 \times 2$ **B: same strip, cambered 6 mm**

The pair is a flat spring blank against a cambered leaf spring, or a flat panel against a roll-formed panel.

Descriptor	Result on this pair, and why
D1, D2	**nearly same** (SNR 1.1 and 1.0). Distances change at second order in the camber (0.24%).
Triangle (A3, D3)	**nearly same** (SNR 2.1 and 2.8). Triangle areas also change only at second order, because tilting a small flat triangle hardly changes its area.
D4	**different** (SNR 50). On the flat strip every tetrahedron is squeezed into the 2 mm thickness. With camber, the ends and the middle differ in height by 6 mm, so volumes grow at *first* order.

The rungs form a pattern. D2 measures 1-D extent, triangles detect departure from a line, and D4 detects departure from a plane. Each one sees at first order what the previous one sees only at second order.

What it still cannot see

All four point-set descriptors are blind to which way the surface *faces*. Shallow surface detail that moves points by a fraction of a millimetre, but tilts the surface strongly, changes none of them noticeably.

3 Which way the surface faces: normal-based descriptors

3.1 SHPS: orientation spectrum

vector length 15

In plain English

Every small piece of surface faces some direction (its normal). Plot all these directions as dots on a unit sphere, each weighted by its area. This "direction cloud" is the Extended Gaussian Image. A cube gives six dots. A cylinder gives a ring around the equator plus two dots at the poles. A ball coats the sphere evenly.
SHPS describes the cloud in a way that does not depend on how the part is turned, much as a sound's frequency spectrum does not depend on when the sound starts. It splits the cloud into

spherical-harmonic "frequencies" $\ell = 2, 4, 6, \ldots$ and keeps only the *amount of energy* in each one.

Mathematics. With area weights $w_t = A_t / A$ and real spherical harmonics Y_ℓ^m,

$$c_{\ell m} = \sum_t w_t \, Y_\ell^m(\hat{\mathbf{n}}_t), \qquad \tilde{P}_\ell = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} c_{\ell m}^2. \tag{11}$$

By the addition theorem $\sum_m Y_\ell^m(\mathbf{a}) Y_\ell^m(\mathbf{b}) = \frac{2\ell+1}{4\pi} \mathcal{P}_\ell(\mathbf{a} \cdot \mathbf{b})$, where $\mathcal{P}_\ell$ is the Legendre polynomial, the same quantity is

$$\tilde{P}_\ell = \sum_s \sum_t w_s w_t \, \mathcal{P}_\ell(\hat{\mathbf{n}}_s \cdot \hat{\mathbf{n}}_t) \ \in [0, 1]. \tag{12}$$

This shows directly that it depends only on the *angles between normals*, so it is rotation invariant. Normals are unaffected by shift and scale. For every closed part $\tilde{P}_0 = 1$ and $\tilde{P}_1 = 0$ (because $\oint \hat{\mathbf{n}} \, dA = \mathbf{0}$), so the feature is $[\tilde{P}_2, \ldots, \tilde{P}_{16}]$.

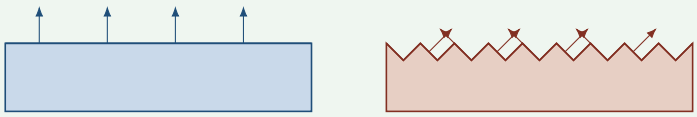
Reference values. Ball: $\tilde{P}_\ell = 0$ for all $\ell \geq 1$. Cube: $\tilde{P}_2 = 0$, $\tilde{P}_4 = 0.583$, $\tilde{P}_6 = 0.125$, $\tilde{P}_8 = 0.516$. Cylinder barrel alone: $\tilde{P}_2 = \frac{1}{4}$.

Algorithm 5 SHPS, computed exactly from the faces

Input: face normals $\hat{\mathbf{n}}_t$, area weights w_t, $L = 16$

1: **for** $m = 0..L$ **do**
2: evaluate normalised associated Legendre $\bar{P}_\ell^m(\hat{\mathbf{n}}_{t,z})$ for $\ell = m..L$ by the standard recurrence
3: $c_{\ell,\pm m} \leftarrow \sum_t w_t \, \bar{P}_\ell^m(\hat{\mathbf{n}}_{t,z}) \, \{\cos m\phi_t, \ \sin m\phi_t\}$ with $\phi_t = \operatorname{atan2}(\hat{\mathbf{n}}_{t,y}, \hat{\mathbf{n}}_{t,x})$
4: $\tilde{P}_\ell \leftarrow \frac{4\pi}{2\ell+1} \sum_m c_{\ell m}^2$
5: **return** $[\tilde{P}_2, \ldots, \tilde{P}_{16}]$

Why it is needed: smooth vs. grooved grip



A: smooth grip (section) **B:** 90° **V-grooves (section, enlarged)**

Both are knobs ∅25 × 60. B has circumferential 90° V-grooves 0.25 mm deep at 0.5 mm pitch (the same effect arises with modelled threads or knurls).

Descriptor	Result on this pair, and why
D1, D2	**nearly same** (SNR 1.8 and 2.5). Surface points move by at most 0.25 mm, which is 0.4% of the part's size.
Triangle (A3, D3)	**nearly same** (SNR 1.6 and 3.2).
D4	**nearly same** (SNR 3.0).
SHPS	**different** (SNR 16). Every piece of the barrel is now tilted by 45°. The single equatorial ring in the direction cloud splits into two rings at ±45° latitude. For the barrel, $\tilde{P}_2$ falls from 1/4 to 1/16.

The reason: position descriptors respond to *how far* points move, and here they move very little. SHPS responds to how much the surface *tilts*, which is large for fine but steep detail.

What it still cannot see

SHPS knows which facing directions occur and how much area each has, but not *where* they occur. Any two parts with the same set of face directions and areas look identical to it, however those faces are arranged.

3.2 NCF: normal correlation over distance

vector length 32

In plain English

Take two random surface points. Note how far apart they are, and whether their surfaces face the same way ($\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j = +1$), opposite ways ($-1$) or at right angles (0). NCF adds up this "facing agreement" separately for each distance.
On a solid block, points close together usually lie on the same face and agree. On a thin-walled hollow part, every piece of the outer skin has a partner on the inner skin a wall-thickness away that faces the *opposite* way, so agreeing and opposing pairs cancel. NCF tells a front-and-back wall from a single solid face, and it also picks up the pitch of repeating features.

Mathematics. Over S random pairs with $\hat{r}_{ij} = \left\|\mathbf{p}_i - \mathbf{p}_j\right\| / D_\varnothing$,

$$g(m) = \frac{1}{S} \sum_{(i,j)} (\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j)\, \mathbb{1}\left[\hat{r}_{ij} \in B_m\right], \qquad m = 1..32. \tag{13}$$

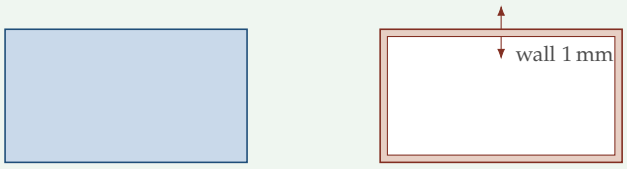
$\hat{r}$ and $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j$ are unchanged by rotation, shift and scale. Two useful facts:

- $\sum_m g(m) = \mathbb{E}[\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j] = \|\mathbb{E}\,\hat{\mathbf{n}}\|^2 = 0$ for every closed part. The total agreement always cancels, and NCF shows *at which distances* agreement and opposition occur.
- $g(m)/p_{\mathrm{D2}}(m)$ is the average normal agreement at distance m, i.e. the correlation function proper. Storing g avoids dividing by near-empty bins.

Algorithm 6 NCF

Input: samples with normals $\{(\mathbf{p}_i, \hat{\mathbf{n}}_i)\}$, $D_\varnothing$, S pairs
1: for each pair: $m \leftarrow \lfloor 32 \left\|\mathbf{p}_i - \mathbf{p}_j\right\| / D_\varnothing \rfloor$; $g[m]$ += $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j$
2: **return** g/S

Why it is needed: solid block vs. sealed hollow box



A: solid block **B: sealed hollow box**

The pair is a solid block against a welded sealed box (a float or tank) with 1 mm walls.

Descriptor	Result on this pair, and why
D1, D2	**nearly same** (SNR 2.2 and 1.9). B's surface is essentially two copies of A's surface, 1 mm apart. A mixture of two nearly identical point clouds has nearly the same distance statistics.
Triangle (A3, D3)	**nearly same** (SNR 1.0 and 2.3). Same reason.
D4	**nearly same** (SNR 3.9). Same reason.
SHPS	**same** (SNR 1.0). The inner skin's normals are the outer ones reversed. For a box the direction cloud is symmetric under reversal, so it is unchanged.
NCF	**different** (SNR 20). For a pair taken from different skins, the normal agreement is the *negative* of what it would be on one skin. Half of B's pairs are such mixed pairs, so $g(m) \approx 0$ at every distance, while A shows a structured profile.

What it still cannot see

NCF, like everything so far, measures positions and directions but not *how the surface is connected*. A narrow slot that nearly cuts a plate in two removes only a few percent of the area, and none of these descriptors react much.

4 How the surface is connected: vibration spectrum

4.1 LBE: Laplace–Beltrami eigenvalues ("ShapeDNA")

vector length 30

In plain English

Imagine the surface is the skin of a drum and tap it. It vibrates in a set of standing-wave patterns, each at its own pitch. The list of pitches depends only on the surface's shape *as measured along the surface*, like an ant walking on it, and not on how it sits in space. Long, narrow paths ring low and compact patches ring high. LBE is the list of the lowest 30 pitches, rescaled so that the part's size drops out.
Because it follows paths *along* the surface, LBE notices cuts, slots and narrow connections that barely change where the surface is.

Mathematics. On the welded mesh, build the cotangent stiffness matrix and the lumped mass matrix:

$$L_{ij} = -\tfrac{1}{2}\big(\cot\alpha_{ij} + \cot\beta_{ij}\big) \;\; (j \text{ adjacent to } i), \qquad L_{ii} = -\sum_{j\neq i} L_{ij}, \qquad M = \operatorname{diag}(\tilde{A}_i). \tag{14}$$

α_{ij}, β_{ij} are the two angles opposite edge ij, and $\tilde{A}_i$ is the area of the mesh around vertex i (mixed Voronoi cell). Solve the generalised eigenproblem

$$L\,\boldsymbol{\phi}_k = \lambda_k\, M\,\boldsymbol{\phi}_k, \qquad 0 = \lambda_0 \le \lambda_1 \le \lambda_2 \le \cdots \tag{15}$$

Eigenvalues scale as $1/\text{length}^2$, so $\hat{\lambda}_k = \lambda_k A$ is scale-free. By Weyl's law $\hat{\lambda}_k \approx 4\pi k$ for *every* surface at large k. The informative part is the deviation from that trend, so the feature is

$$F_{\text{LBE}} = \Big[\log\frac{\hat{\lambda}_k}{4\pi k}\Big]_{k=1}^{30}. \tag{16}$$

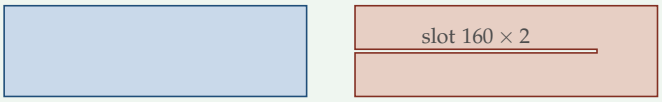
The spectrum depends only on lengths and angles measured on the surface, so it is unchanged by rotation, shift and mirroring. If the part has several separate bodies, one zero eigenvalue per body is skipped. Meshes with many sliver triangles should first be made intrinsic-Delaunay (edge flips) so that all cotangent weights are non-negative.

Algorithm 7 LBE

Input: welded manifold mesh (intrinsic-Delaunay flipped), $K = 30$

1: assemble sparse L and diagonal M from per-triangle cotangents
2: $\beta_0 \leftarrow$ number of connected bodies
3: solve for the $K + \beta_0$ smallest eigenvalues with shift-invert Lanczos (`eigsh`, $\sigma \approx -10^{-8}$)
4: drop the first β_0; $\hat{\lambda}_k \leftarrow \lambda_k A$
5: **return** $\big[\log(\hat{\lambda}_k/4\pi k)\big]_{k=1}^{K}$

Why it is needed: plain plate vs. slotted plate



A: plate $200 \times 60 \times 2$ **B: same plate, slot from one end**

The pair is a flat plate against the same plate with a saw-cut slot, which turns it into a fork or flexure.

Descriptor	Result on this pair, and why
D1–D4	**nearly same**. The slot removes 2.7% of the face area. By the small-feature principle (4), no point-set histogram can move by more than $2k \times 0.027$.
SHPS	**nearly same**. The slot walls add a small area facing $\pm y$ (the plate is thin).
NCF	**nearly same**. Same small area.
LBE	**different**. The two prongs can now swing independently, and a path along the surface from one prong tip to the other must go around the end of the slot. On the plate face (a 1 mm finite-difference Neumann computation), λ_1 drops to $0.33\times$ and λ_2 to $0.25\times$ their unslotted values: a 3–4-fold change from 2.7% of the area.

What it still cannot see

LBE depends only on intrinsic geometry, so it is blind to anything a mirror or a bend can undo. A boss and a pocket of the same size are exact mirror images of each other through the plate face, and give *identical* spectra. It also barely notices local edge treatment such as sharp versus chamfered edges.

5 Local shape: edges and curvature

5.1 DAD: dihedral angle distribution

vector length 36 + 2

In plain English

Look at every edge where two flat mesh faces meet and measure the fold angle between them. 180° means flat (no real edge), 90° a sharp outside corner like a box edge, and 270° a sharp inside corner like the floor of a slot. DAD is the histogram of fold angles over the real creases, weighted by edge length.

Two extra numbers go with it: the total length of creases, and the total bending (how much the surface turns overall, summed along all edges). The total bending is the same for a sharp edge and for a rounded one. It tells "how much the surface turns", while the histogram tells "how sharply".

Mathematics. For an edge e of length ℓ_e shared by faces i, j with outward normals $\hat{\mathbf{n}}_i, \hat{\mathbf{n}}_j$ and centres $\mathbf{b}_i, \mathbf{b}_j$,

$$\sigma_e = \operatorname{sign}\big((\mathbf{b}_j - \mathbf{b}_i) \cdot \hat{\mathbf{n}}_i\big), \qquad \theta_e = \pi + \sigma_e \arccos(\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j) \in [0, 2\pi]. \tag{17}$$

$\sigma_e < 0$ means the neighbour bends away (a convex edge, $\theta < \pi$), and $\sigma_e > 0$ means it bends up (a concave edge, $\theta > \pi$). Only creases with $|\theta_e - \pi| > \tau$ are histogrammed, with $\tau = 20°$ chosen above the exporter's facet angle. This way smoothly curved faces, which the STL splits into many

nearly flat facets, do not count as edges.

$$F_{\mathrm{DAD}}(b) = \frac{\sum_{e:\,\theta_e \in b,\,|\theta_e-\pi|>\tau} \ell_e}{\sum_{e:\,|\theta_e-\pi|>\tau} \ell_e} \ \text{(36 bins of 10°)}, \qquad \Lambda = \frac{\sum_{\text{creases}} \ell_e}{\sqrt{A}}, \qquad \mathcal{H} = \frac{\frac{1}{2}\sum_e (\pi-\theta_e)\,\ell_e}{\sqrt{A}}. \quad (18)$$

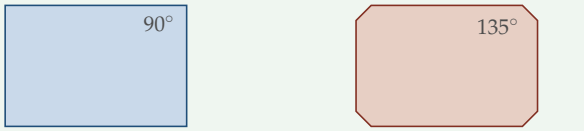
$\mathcal{H}$ is the discrete total mean curvature (Steiner's formula). It does not depend on how finely a round was tessellated. Angles between normals, and lengths divided by $\sqrt{A}$, are unchanged by rotation, shift and scale.

Algorithm 8 DAD

Input: welded mesh with edge → (face, face) map, τ

1: **for** each edge with exactly two faces **do**
2: compute θ_e and ℓ_e; add $(\pi-\theta_e)\ell_e/2$ to $\mathcal{H}$
3: **if** $|\theta_e-\pi|>\tau$ **then** add ℓ_e to bin of θ_e and to Λ
4: **return** L1-normalised histogram, $\Lambda/\sqrt{A}$, $\mathcal{H}/\sqrt{A}$

Why it is needed: sharp edges vs. chamfered edges



A: block, sharp edges **B: chamfered** $0.5 \times 45°$

The pair is a $60 \times 40 \times 30$ block as sawn against the same block deburred with $0.5 \times 45°$ chamfers on all 12 edges.

Descriptor	Result on this pair, and why
D1–D4	**nearly same**. The chamfers change a strip about 0.5 mm wide along each edge, roughly 5% of the area, so every histogram moves by only a few percent.
SHPS, NCF	**nearly same**. About 3% of the area now faces diagonally.
LBE	**nearly same**. The metric changes only in thin strips.
DAD	**different**. In A *all* crease length sits in the 90° bin. In B it sits in the 135° bin, with two creases per chamfer. The histograms do not overlap, which is the maximum possible L1 distance of 2. If the edges are filleted instead, the creases vanish ($\Lambda \to 0$) while $\mathcal{H}$ stays the same, because the total turning is unchanged.

What it still cannot see

DAD sees only folds at edges. Smoothly curved faces have no creases, so a convex round boss and a concave round pocket of the same size, each with one convex and one concave rim, give *identical* DAD.

5.2 Curvature descriptor: shape index and curvedness

vector length 97

In plain English

At any point of a smooth face, the surface bends by different amounts in different directions. The strongest and weakest bends are the two *principal curvatures* $\kappa_1 \geq \kappa_2$. Each is 1/radius, positive when the surface bulges outward (convex) and negative when it dishes inward (concave). Two numbers summarise the pair:

- **Shape index** SI answers *what kind* of shape it is: -1 cup (inside of a ball), $-\frac{1}{2}$ trough (inside of a bore), 0 saddle, $+\frac{1}{2}$ ridge (outside of a shaft), $+1$ cap (outside of a ball).
- **Curvedness** CV answers *how tightly* it bends: 1/radius, so a R1 fillet is five times more curved than a R5 fillet.

The descriptor is a 2-D histogram of (SI, CV) over the curved part of the surface, plus the share of the surface that is flat.

Mathematics. *Principal curvatures at a fixed scale.* At each surface sample $\mathbf{p}$, take the neighbours within radius $r = 0.02\,D_{\varnothing}$, express them in a local frame whose z-axis is the outward normal, and fit $z = ax^2 + bxy + cy^2 + dx + ey$ by least squares. Then

$$\kappa_{1,2} = \text{eigenvalues of } \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}, \qquad \kappa_1 \geq \kappa_2. \tag{19}$$

Using a fixed physical neighbourhood, rather than the raw STL triangles, makes the result independent of how the exporter tessellated the part. Samples whose neighbourhood contains a crease (neighbour normals differing by more than $\tau = 20^\circ$) are left out, because sharp edges are DAD's job. The curvature descriptor therefore describes the *smooth* faces only.

Shape index and curvedness.

$$\mathrm{SI} = \frac{2}{\pi}\,\mathrm{atan2}\left(\kappa_1 + \kappa_2,\ \kappa_1 - \kappa_2\right) \in [-1, 1], \qquad \mathrm{CV} = \sqrt{\tfrac{1}{2}(\kappa_1^2 + \kappa_2^2)}, \qquad \widehat{\mathrm{CV}} = \mathrm{CV}\,\sqrt{A/4\pi}. \tag{20}$$

The atan2 form is well defined everywhere except at perfectly flat points, where $\mathrm{CV} = 0$. The feature is an area-weighted histogram with 12 SI bins $\times$ 8 log-spaced $\widehat{\mathrm{CV}}$ bins over the smooth, curved samples ($\widehat{\mathrm{CV}} > \varepsilon$), plus one number for the flat share of the smooth area: $12 \times 8 + 1 = 97$ values. SI is a ratio (scale-free), $\widehat{\mathrm{CV}}$ is normalised by $\sqrt{A}$, and both are unchanged by rotation and shift.

The four classical curvature views are read off the same histogram. (SI, CV) are polar coordinates of the point (κ_1, κ_2): rotate the plane by 45° to $u = (\kappa_1 + \kappa_2)/\sqrt{2}$, $v = (\kappa_1 - \kappa_2)/\sqrt{2}$, and then the radius is $\sqrt{2}$ CV and the angle is $\frac{\pi}{2}$ SI. Hence

$$K = \kappa_1\kappa_2 = -\,\mathrm{CV}^2\cos(\pi\,\mathrm{SI}), \qquad H = \tfrac{1}{2}(\kappa_1 + \kappa_2) = \mathrm{CV}\sin\!\left(\tfrac{\pi}{2}\,\mathrm{SI}\right). \tag{21}$$

View	What it isolates	Pair it separates
Gaussian curvature K (GCD)	doubly curved vs. single-curved	ball-nose end ($K > 0$) vs. cylinder or cone ($K = 0$) vs. saddle-shaped fillet ($K < 0$)
Mean curvature H (MCD)	convex vs. concave	outside of a boss ($H > 0$) vs. inside of a bore ($H < 0$)
Shape index SI (SID)	type of shape, independent of size	ridge vs. trough vs. cap vs. cup vs. saddle
Curvedness CV (CVD)	size of rounding, independent of type	R1 fillet vs. R5 fillet (both SI $= +\frac{1}{2}$)

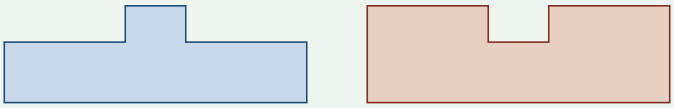
The joint histogram keeps all four views *and* their correlation, e.g. "tight concave" vs. "gentle concave". The ratio κ_2/κ_1 is a function of SI alone, so it needs no separate block.

Algorithm 9 Curvature descriptor

Input: samples with normals $\{(\mathbf{p}_i, \hat{\mathbf{n}}_i)\}$, $D_\varnothing$, total area A

build a kd-tree on $\{\mathbf{p}_i\}$; $r \leftarrow 0.02\, D_\varnothing$
for each sample i **do**
  if some neighbour normal differs from $\hat{\mathbf{n}}_i$ by more than τ **then** skip i (crease region)
  neighbours within $r \rightarrow$ local frame $(\mathbf{e}_x, \mathbf{e}_y, \hat{\mathbf{n}}_i) \rightarrow$ least-squares quadric fit
  $\kappa_{1,2} \leftarrow$ eigenvalues of the Hessian; SI_i, CV_i as above
flat fraction $\leftarrow$ share of samples with $\widehat{\mathrm{CV}} < \varepsilon$
return L1-normalised 12×8 histogram of $(\mathrm{SI}, \log \widehat{\mathrm{CV}})$ over curved samples, flat fraction

Why it is needed: boss vs. pocket



A: ∅20 × 10 **boss** **B:** ∅20 × 10 **pocket**

The pair is a 100 × 100 × 20 plate with a round boss on top, against the same plate with a round pocket of the same size milled in.

Descriptor	Result on this pair, and why
D1–D4	**nearly same**. The feature is about 2% of the area and does not change $D_\varnothing$.
SHPS	**same** (exact). The boss wall faces outward in all horizontal directions and the pocket wall faces inward in all horizontal directions, which is the same set of directions with the same area. The $+z$ area is also equal: the boss top replaces the covered plate area, and the pocket floor replaces the removed plate area.
NCF	**nearly same**. Only a second-order distance weighting differs.
LBE	**same** (exact). The pocket is the boss mirrored through the plate face, so every length along the surface is the same.
DAD	**same** (exact). Both have one convex 90° rim and one concave 270° rim, each 20π mm long.
Curvature	**different**. The boss wall is a convex cylinder ($\mathrm{SI} = +\frac{1}{2}$) and the pocket wall a concave one ($\mathrm{SI} = -\frac{1}{2}$). Everything else is identical, and the plate's flat faces and creases are excluded or counted as flat. So the whole curved part of the histogram moves from the $+\frac{1}{2}$ bin to the $-\frac{1}{2}$ bin, the maximum possible difference.

What it still cannot see

Curvature is local. It does not know *where* a curved face sits relative to the rest of the part. A stepped shaft whose middle section is offset (eccentric) has exactly the same cylinders, and so exactly the same curvature histogram, as the concentric shaft.

6 CAD-specific geometry and topology

The descriptors so far are general 3-D shape descriptors. The remaining six encode quantities that mechanical design produces on purpose: turning axes and diameters, through-holes, bolt circles, indexing patterns, datum planes and wall thickness.

6.1 CRS-R: concentric ring profile

vector length 128

In plain English

Find the part's own axis: for a turned part, the axis it was turned about. Then measure the distance of every surface point from that axis and histogram it. Each cylindrical section of a shaft shows up as a sharp spike at its radius. Shoulders and end faces show up as ramps between radii. The profile is a direct read-out of the part's *diameter table*, and it shows whether all the diameters share one axis.

Mathematics. *Axis.* Compute the covariance of the samples, $C = \mathrm{cov}\{\mathbf{p}_i\}$, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ and eigenvectors $\mathbf{e}_a$. A body of revolution has two equal eigenvalues, and the remaining eigenvector is its axis. So

$$s = \arg\min_a \frac{|\lambda_b - \lambda_c|}{\lambda_1} \qquad (\{a,b,c\} = \{1,2,3\}). \tag{22}$$

Profiles. The distance to an axis through $\mathbf{c}$ along $\mathbf{e}_a$ is $\rho_i^{(a)} = \|(\mathbf{p}_i - \mathbf{c}) \times \mathbf{e}_a\| / D_\varnothing$. The feature is the area-weighted histogram of $\rho^{(s)}$ with 64 bins (fine bins, because diameters matter), followed by the two other axes with 32 bins each, in eigenvalue order.

Exact model for a turned part. A barrel of radius r_i and length h_i projects to a single radius, and a flat annular face covers a band of radii with area element $2\pi r\, dr$. Hence

$$R(r) = \sum_i 2\pi r_i h_i\, \delta(r - r_i) \;+\; 2\pi r\, m(r), \tag{23}$$

where $m(r)$ is the number of flat faces covering radius r. Peak positions give the radii, and peak masses divided by $2\pi r_i$ give the step lengths.

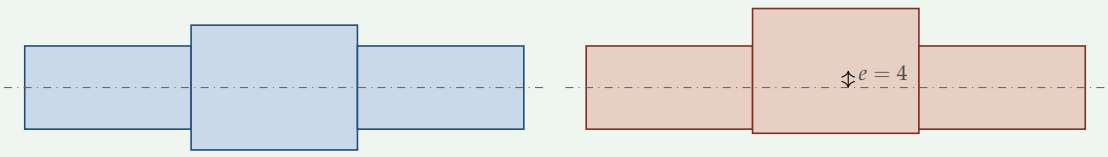
Invariance. A distance to an axis does not depend on the axis sign or on the choice of axes perpendicular to it. For a body of revolution, every perpendicular axis gives the same profile, so the arbitrary eigenvector choice in the degenerate plane is harmless.

Algorithm 10 CRS-R

Input: samples $\{\mathbf{p}_i\}$ with area weights, $D_\varnothing$

1: $\mathbf{c} \leftarrow$ mean; $(\lambda, \mathbf{e}) \leftarrow$ eigen-decomposition of $\mathrm{cov}\{\mathbf{p}_i\}$; choose s
2: **for** a in (s, then the other two by eigenvalue) **do**
3: $\quad \rho \leftarrow \|(\mathbf{p} - \mathbf{c}) \times \mathbf{e}_a\| / D_\varnothing$; histogram (64 bins for s, 32 otherwise)
4: **return** concatenated, L1-normalised profiles

Why it is needed: concentric vs. eccentric stepped shaft



A: concentric ∅20/∅30/∅20 **B: middle section offset 4 mm**

The pair is a plain stepped shaft (∅20 × 40, ∅30 × 40, ∅20 × 40) against an eccentric shaft (cam shaft) with the middle journal offset by $e = 4$ mm. Measured with $n = 2 \times 10^5$ samples.

Descriptor	Result on this pair, and why
D1	**nearly same** (SNR 4.9). It notices that the centre shifted slightly, but the change looks like a small change of proportions.
D2, triangle, D4	**nearly same** (SNR 1.2, 1.4/2.0, 2.3).
SHPS	**same** (SNR 0.8). A cylinder's normals form the same ring wherever its axis is.
NCF	**nearly same** (SNR 1.3).
LBE	**nearly same**. Only the two shoulder rings change shape.
DAD	**same** (exact). The same circles meet at the same 90° and 270° folds.
Curvature	**same** (exact). The same three cylinders are present.
CRS-R	**different** (SNR 32). In A every barrel is a single sharp spike at its radius. In B the barrels no longer share one axis, so each spike spreads into a band about twice as wide as its offset (the middle one covers roughly 13–17 mm instead of exactly 15 mm).

What it still cannot see

CRS-R measures *amounts of area* at each radius. It cannot tell whether a hole goes through or stops short, because the walls of both sit at the same radii.

6.2 T1: genus and body count (topology)

vector length 4

In plain English

Count the holes that go *all the way through* the part, the *genus*. A washer has 1, a plate with four bolt holes has 4, and a block with a blind pocket has 0. T1 also counts separate bodies and sealed internal cavities. These are whole numbers: they do not change when the part is stretched or bent, and they jump by exactly one when a hole breaks through.

Mathematics. For each connected surface component c of the welded mesh, with V_c vertices, E_c edges, F_c faces and b_c boundary loops (0 for a watertight part), the Euler–Poincaré formula gives

$$\chi_c = V_c - E_c + F_c = 2 - 2g_c - b_c \quad \Longrightarrow \quad g_c = \tfrac{1}{2}(2 - \chi_c - b_c). \tag{24}$$

The signed volume $V_c = \frac{1}{6}\sum_{t\in c} \mathbf{x}_0 \cdot (\mathbf{x}_1 \times \mathbf{x}_2)$ is positive for an outer skin and negative for the wall of an internal cavity. The feature is

$$F_{\mathrm{T1}} = \big[\#\text{bodies},\ \log(1+g),\ \#\text{cavities},\ \textstyle\sum_c b_c\big], \qquad g = \textstyle\sum_c g_c. \tag{25}$$

The last entry is a mesh-quality flag and should be 0. Topology is unchanged by any continuous deformation, so T1 is invariant under far more than rotation, shift and scale.

Algorithm 11 T1

Input: welded mesh with edge → faces map

1: split faces into connected components (union–find over shared edges)
2: **for** each component **do** count V_c, E_c, F_c; trace boundary loops b_c; $g_c \leftarrow (2 - \chi_c - b_c)/2$; sign of V_c
3: **return** [bodies ($V_c > 0$), $\log(1 + \sum g_c)$, cavities ($V_c < 0$), $\sum b_c$]

Why it is needed: through-holes vs. blind holes



A: 4 through-holes ∅3 **B: 4 blind holes, 0.2 mm short**

The pair is a $100 \times 80 \times 10$ plate with four ∅3 through-holes, against the same plate whose holes stop 0.2 mm before the bottom face.

Descriptor	Result on this pair, and why
D1–D4, SHPS, NCF, CRS-R	**nearly same**. The holes are under 1% of the area, and the two versions differ only in a few tiny discs.
LBE	**nearly same**. A 3 mm tunnel barely changes the low vibration modes of a 100 mm plate.
DAD	**nearly same**. One rim per hole changes from a convex to a concave fold, about 38 mm out of about 835 mm of creases ($\approx 4.5\%$).
Curvature	**nearly same**. The hole walls are the same concave cylinders; the blind floors are flat.
T1	**different**. Genus $g = 4$ against $g = 0$.

Every earlier descriptor measures an *amount* (area, length, curvature, pitch), and amounts change only a little when a little material changes. Topology is a *count*: breaking through the last 0.2 mm changes it by a whole unit per hole.

What it still cannot see

T1 counts holes but says nothing about *where* they are or how big they are. Six holes on a small bolt circle and six on a large one give the same genus.

6.3 PHD: persistent homology (where the holes are)

vector length 77

In plain English

Fill the part with "rising water" measured outward from its axis. At level t, keep only the surface closer than t to the axis. As t grows, pieces appear, merge, and loops form around holes and later close. Each hole produces a loop that is *born* at one level and *dies* at another. The pair (birth, death) says where the hole sits and how wide it is, whatever its area. All the pairs, drawn as points on a plane, form the persistence diagram, which is then converted to a small fixed-size image.

Mathematics. Filtration function on the vertices: the cylindrical radius $f(\mathbf{p}) = \|(\mathbf{p} - \mathbf{c}) \times \mathbf{e}_s\| / D_\varnothing$ about the axis $\mathbf{e}_s$ of §6.1. Each edge and triangle takes the largest f of its vertices (lower-star filtration), giving nested complexes $\mathcal{M}_t = \{\sigma : f(\sigma) \le t\}$. *Extended* persistence (an upward sweep, then a downward relative sweep) gives every homology class a finite (birth, death) pair, including the $2g$ loops that belong to the through-holes. It yields three diagrams: PD_0 (pieces), PD_1 (loops) and Ext_1 (hole loops).

Persistence image. Map each point to $(b,\ p = d - b)$ and sum weighted Gaussians on a 5×5 grid:

$$I(x,y) = \sum_{(b,p)} w(p)\, \exp\Big(- \frac{(x-b)^2 + (y-p)^2}{2\sigma^2} \Big), \qquad w(p) = \min(1, p/p_{\max}). \tag{26}$$

With three diagrams and two scalars (the number of loops with persistence above 0.02, and the total persistence), the feature has $3 \times 25 + 2 = 77$ values.

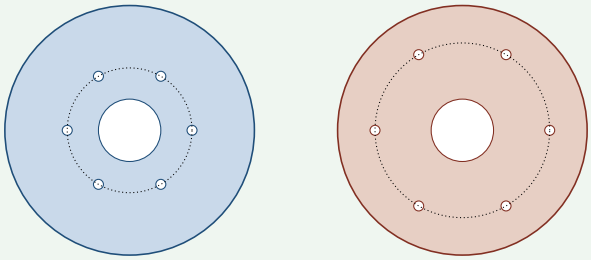
Stability. For two filtration functions, the bottleneck distance between their diagrams is at most $\|f - g\|_\infty$. Mesh noise of size δ therefore moves the diagram by at most δ, while a real hole produces a point that noise cannot remove.

Algorithm 12 PHD

Input: welded mesh, axis $\mathbf{e}_s$, centre $\mathbf{c}$, $D_\varnothing$

1: $f_v \leftarrow \|(\mathbf{x}_v - \mathbf{c}) \times \mathbf{e}_s\| / D_\varnothing$ for all vertices
2: build the lower-star filtration on vertices, edges and triangles
3: compute extended persistence (e.g. GUDHI `extend_filtration` + `persistence`)
4: **return** persistence images of PD_0, PD_1, Ext_1, plus the two scalars

Why it is needed: same holes, different bolt circle



A: $6 \times \varnothing 4$ **on R30** **B:** $6 \times \varnothing 4$ **on R42**

The pair is a ∅120 flange (bore ∅30, 12 mm thick) with six ∅4 holes, on a R30 bolt circle in A and a R42 bolt circle in B.

Descriptor	Result on this pair, and why
T1	**same** (exact). $g = 7$ in both (six holes plus the bore).
DAD, curvature	**same** (exact). Identical holes give identical rims and walls.
SHPS	**same** (SNR 0.8).
D1–D4, NCF	**nearly same** (SNR 1.9, 1.7, 1.1/2.3, 2.2, 0.9). With ten times more samples D1 rises to about 4: it notices that 3% of the area moved outward, but it cannot distinguish that from, say, a slightly smaller bore.
CRS-R	**nearly same** (SNR 1.6). The hole walls are too little area to move the profile.
LBE	**nearly same**. Small holes, small change.
PHD	**different**. The loop of each hole lives between the critical values $R_b - r_h$ and $R_b + r_h$ of the radial sweep: 28–32 mm in A, 40–44 mm in B. The hole points move across the diagram, independent of how little area the holes have.

What it still cannot see

With the radial sweep, PHD knows how far each hole is from the axis, but not how the holes are *spaced around* it. Six holes at equal 60° pitch and six holes bunched in two groups give the same diagram.

6.4 CRS-A: angular pattern spectrum

vector length 128

In plain English

Walk once around the part's axis and record how much surface lies in each direction, separately for a few rings (radius bands). A regular pattern of N features (gear teeth, splines, bolt holes,

spokes) repeats every $360^\circ / N$ and produces a strong peak at harmonic N of this angular profile. Only the *height* of each harmonic is kept, not its phase, so it does not matter where the part starts or how the part is turned about its axis.

Mathematics. With $\mathbf{e}_s$ from §6.1 and the other two eigenvectors $\mathbf{e}_b, \mathbf{e}_c$, compute the azimuth $\varphi_i = \operatorname{atan2}\big((\mathbf{p}_i - \mathbf{c}) \cdot \mathbf{e}_b,\ (\mathbf{p}_i - \mathbf{c}) \cdot \mathbf{e}_c\big)$. Split the radius range of $\rho^{(s)}$ into 4 equal bands. In band q, histogram φ into $N_\varphi = 256$ bins to get $a_q(j)$, then

$$\hat{a}_q(m) = \sum_{j=0}^{N_\varphi - 1} a_q(j)\, e^{-2\pi i\, mj / N_\varphi}, \qquad F_{\text{CRS-A}} = \left[\frac{|\hat{a}_q(m)|}{\hat{a}_q(0)} \right]_{q=1..4,\ m=1..32}. \tag{27}$$

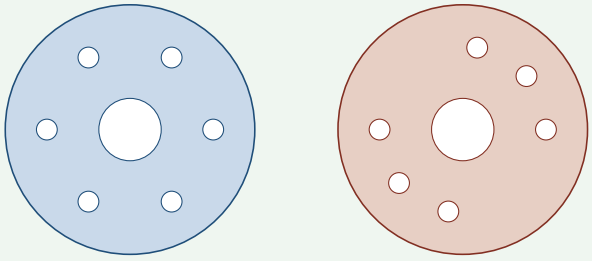
Rotating the part about $\mathbf{e}_s$ shifts φ by a constant, which multiplies $\hat{a}_q(m)$ by a unit phase, so its magnitude is unchanged. Flipping $\mathbf{e}_b$ or $\mathbf{e}_c$ maps $\varphi \to -\varphi$ or $\varphi + \pi$, and magnitudes are again unchanged. Detecting N features requires $m_{\max} \geq N$. Because patterns occupy narrow bands, CRS-A needs more samples than the other blocks ($n \approx 2 \times 10^5$).

Algorithm 13 CRS-A

Input: samples, $\mathbf{c}$, $\mathbf{e}_s, \mathbf{e}_b, \mathbf{e}_c$, 4 bands, $N_\varphi = 256$, $M = 32$

1: **for** each band q **do**
2: histogram φ of the samples in the band; FFT; keep $|\hat{a}_q(1..M)| / \hat{a}_q(0)$
3: **return** the $4 \times M$ magnitudes

Why it is needed: equal pitch vs. grouped holes



A: $6 \times 60^\circ$ **equal pitch** **B: two groups of 3 at** 40°

The pair is the same ∅120 flange with six ∅10 holes on R40, equally spaced in A and grouped for a two-piece clamp in B. Measured with $n = 2 \times 10^5$ samples.

Descriptor	Result on this pair, and why
D1	**same** (exact; SNR 0.9). B is symmetric under a half turn, so the centre stays put, and every surface point keeps its distance from it.
D2–D4	**nearly same** (SNR 1.1, 1.0/1.0, 0.9). Only hole-to-hole distances change, and holes are a small share of the area.
SHPS	**same** (exact; SNR 0.8). Identical holes have identical wall normals.
NCF	**nearly same** (SNR 1.1).
CRS-R	**same** on the flange axis (radii unchanged). The two in-plane profiles shift slightly (SNR 1.8).
LBE	**nearly same**.
DAD, curvature, T1, PHD	**same** (exact). The holes are identical and sit at the same radius.
CRS-A	**different** (SNR 4.3). A has a peak at $m = 6$, since $\left\lvert \sum_k e^{i6\theta_k} \right\rvert = 6$. In B the sixth harmonic cancels exactly ($\sum_k e^{i6\theta_k} = 0$), and harmonic $m = 2$ appears instead.

What it still cannot see

CRS-A describes rotational patterns only. It cannot tell whether a set of flat faces is *coplanar*, for example whether all mounting feet of a housing touch the same plane.

6.5 PAF: planar face families

vector length 31

In plain English

Machined parts are largely made of flat faces, many of them parallel: top, bottom, pads, steps, pockets. PAF groups all flat faces into *families* that point the same way (or exactly the opposite way), and for each family counts how many distinct planes it contains, how much area it carries, and how evenly the planes are spaced. It also reports the share of the surface that is flat ("prismatic-ness"), which separates machined blocks from free-form castings.

Mathematics. *Global normal moments.* With area weights w_t,

$$T_2 = \sum_t w_t\, \hat{\mathbf{n}}_t \hat{\mathbf{n}}_t^{\top}, \quad \Lambda_1 \geq \Lambda_2 \geq \Lambda_3 \text{ its eigenvalues}, \quad c_l = \Lambda_1 - \Lambda_2,\ c_p = 2(\Lambda_2 - \Lambda_3),\ c_s = 3\Lambda_3, \tag{28}$$

so that $c_l + c_p + c_s = 1$ (plate-like, barrel-like, box- or ball-like). Also $I_4 = \sum_{s,t} w_s w_t (\hat{\mathbf{n}}_s \cdot \hat{\mathbf{n}}_t)^4 = \|T_4\|_F^2$ with $T_4 = \sum_t w_t \hat{\mathbf{n}}_t^{\otimes 4}$, which equals $\frac{1}{3}$ for a cube and $\frac{1}{5}$ for a ball.

Families. Sort planar faces by area. Assign each face to the first family direction $\mathbf{m}_s$ with $|\hat{\mathbf{n}}_t \cdot \mathbf{m}_s| \geq \cos 0.25^\circ$, or start a new family. CAD planar faces have exact normals, so this resolves draft angles down to about 0.5°. Family share $\alpha_s = \sum_{t \in s} w_t$; prismatic-ness $\Pi = \sum_s \alpha_s$.

Planes within a family. The offsets $h_t = \mathbf{b}_t \cdot \mathbf{m}_s / D_\varnothing$ are clustered in 1-D (tolerance 10^{-4}) into N_s distinct planes. The gaps Δ_k between successive planes give a regularity $\mathrm{CV}_s = \sigma(\Delta)/\mu(\Delta)$ when $N_s \geq 3$.

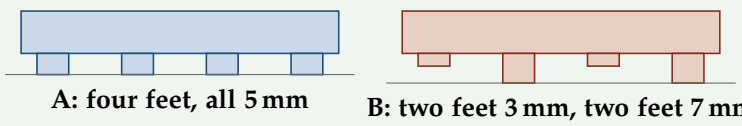
Feature. $[c_l, c_p, c_s, I_4, H, N_{\text{fam}}/16, \Pi]$ plus, for the 6 largest families, $[\alpha_s,\ N_s/10,\ \mathrm{CV}_s,\ \angle(\mathbf{m}_s, \mathbf{m}_1)/90^\circ]$: $7 + 24 = 31$ values. H is the entropy of the normal directions on an equal-area icosphere grid. Everything is built from angles between normals and from offset *differences* divided by $D_\varnothing$, so it is invariant.

Algorithm 14 PAF

Input: faces with normals, areas, centres; tolerance 0.25°

1: $T_2, T_4 \leftarrow$ area-weighted sums; Westin coefficients, I_4, entropy H
2: group coplanar-normal faces into families greedily by area
3: **for** each of the 6 largest families **do** cluster offsets into planes; N_s, gaps, CV_s, angle to family 1
4: **return** the 31-value vector (families sorted by area, zero-padded)

Why it is needed: coplanar feet vs. stepped feet



A: four feet, all 5 mm **B: two feet 3 mm, two feet 7 mm**

The pair is a $150 \times 100 \times 15$ base with four 20×20 mounting feet. In A the feet are all 5 mm tall and coplanar, so the part sits flat. In B two feet are 3 mm and two are 7 mm: the same average height, but the part rocks.

Descriptor	Result on this pair, and why
Curvature	**same** (exact). Everything is flat.
SHPS	**same** (exact). Side-wall area per direction is $4 \times 20 \times 5 = 400$ in A and $2 \times 20 \times 3 + 2 \times 20 \times 7 = 400$ in B.
DAD	**same** (exact). Vertical edge length is $16 \times 5 = 80$ in A and $8 \times 3 + 8 \times 7 = 80$ in B, with the same rims and angles.
T1, PHD	**same**. No holes; the same components.
D1–D4, NCF, CRS-R/A, LBE	**nearly same**. 4% of the area moves by 2 mm on a 180 mm part.
PAF	**different**. The vertical family has 3 distinct planes in A (top, bottom, feet) and 4 in B. The gap pattern changes from $\{15, 5\}$ to $\{15, 3, 4\}$ mm (relative to $D_\varnothing$). The coplanar datum is directly visible.

What it still cannot see

PAF, like every descriptor before it, looks at the surface only. It cannot tell how much material lies *behind* a surface: a thin-walled cover and a thick-walled cover with the same outer shape look alike.

6.6 MTS: wall thickness spectrum

vector length 40

In plain English

From each surface point, shoot a small bundle of rays straight into the material and measure how far they travel before they come out the other side. That distance is the local wall thickness. MTS is the histogram of wall thickness, on a log scale because thicknesses range from sheet gauges to solid sections, plus a few summary numbers such as the share of thin walls. Wall thickness decides which manufacturing process can make a part (sheet metal, moulding, casting, machining from solid), and no surface-only measurement captures it directly.

Mathematics (Shape Diameter Function). From sample $\mathbf{p}_i$ with outward normal $\hat{\mathbf{n}}_i$, cast $c = 16$ rays with directions $\mathbf{d}_{ij}$ inside a cone of half-angle 30° about $-\hat{\mathbf{n}}_i$, starting at $\mathbf{p}_i - \epsilon\hat{\mathbf{n}}_i$ with $\epsilon = 10^{-6} D_\varnothing$. With τ_{ij} the distance to the first hit,

$$w_i = \operatorname*{median}_j \left\{ \tau_{ij} \cos \angle(\mathbf{d}_{ij}, -\hat{\mathbf{n}}_i) \right\}, \qquad \hat{w}_i = \frac{w_i}{D_\varnothing} \in (0, 1]. \tag{29}$$

The median, taken after rejecting values more than one standard deviation from the mean, suppresses the rays that clip a nearby edge. The radius r of the largest ball that touches the surface at $\mathbf{p}_i$ from inside satisfies $r \le \tau_i/2$, with equality for parallel walls. The feature is

$$F_{\mathrm{MTS}} = \Big[\mathrm{hist}_{32} \left(\log_{10} \hat{w}; [-3, 0] \right),\ \mu, \sigma, \gamma_1, Q_1, Q_2, Q_3 \text{ of } \log_{10} \hat{w},\ f_{\mathrm{thin}},\ f_{\mathrm{miss}} \Big]. \tag{30}$$

Here $f_{\mathrm{thin}} = \Pr[\hat{w} < 0.01]$, walls thinner than 1% of the part's size. f_{miss} is the share of rays that never hit anything, which is 0 for a watertight part and serves as a mesh-quality flag. Thickness is a distance inside the part divided by $D_\varnothing$, so it is invariant.

Algorithm 15 MTS

Input: mesh, samples with normals, $c = 16$, cone 30°

1: build a bounding-volume hierarchy (e.g. Embree) over the triangles
2: **for** each sample **do** cast c rays into the material; keep first-hit distances; $w_i \leftarrow$ robust median
3: **return** log-thickness histogram, moments, f_{thin}, f_{miss}

Why it is needed: thin-walled vs. thick-walled cover



A: moulded cover, wall 1.5 mm



B: same outside, wall 3 mm

The pair is a 150 × 100 × 40 moulded cover with the same outer surface, with a 1.5 mm wall in A and a 3 mm wall in B ($D_{\varnothing} \approx 184$ mm).

Descriptor	Result on this pair, and why
D1–D4	**nearly same**. The inner surface moves by 1.5 mm, under 1% of the size. Pairs across the wall fall into D2's first bin (< 5.7 mm) in both.
SHPS	**nearly same**. The inner normals are the reversed outer normals in both.
NCF	**nearly same**. Both have opposing pairs across the wall, and both spacings fall inside the first distance bin.
LBE	**nearly same**. The inner skin is only slightly smaller.
DAD	**same**. Same rim folds, nearly the same lengths.
Curvature	**nearly same**. The inner radii of curvature change by $t/R \approx 2\%$.
CRS-R/A, T1, PHD, PAF	**same** or **nearly same**. Same topology, same planes, nearly the same radial and angular profiles.
MTS	**different**. The thickness spike sits at $\log_{10}(1.5/184) = -2.09$ in A and $\log_{10}(3/184) = -1.79$ in B. That is 0.3 decades apart, more than three bins of the log histogram. f_{thin} (walls under $0.01 D_{\varnothing} = 1.84$ mm) goes from 1 to 0.

The thickness difference is the one that decides whether the cover can be injection-moulded without sink marks, and only MTS measures it directly.

7 Putting it together

7.1 Summary: what each descriptor adds

#	Descriptor	Measures	Pair that only this rung separates	Length
1	D1	distance from centre	ball vs. cube (baseline)	32
2	D2	distance between surface points	evenly spaced vs. paired spokes	32
3	Triangle (A3 + D3)	departure from a line	straight vs. bowed rod	64
4	D4	departure from a plane; handedness	flat vs. cambered strip; part vs. mirror image	33
5	SHPS	facing directions	smooth vs. grooved grip	15
6	NCF	facing agreement vs. distance	solid block vs. sealed hollow box	32
7	LBE	connectivity along the surface	plain vs. slotted plate	30
8	DAD	sharpness of edges	sharp vs. chamfered block	38
9	Curvature (SI + CV)	convex/concave, type and size of rounds	boss vs. pocket	97
10	CRS-R	diameters about the part axis	concentric vs. eccentric shaft	128

#	Descriptor	Measures	Pair that only this rung separates	Length
11	T1	number of through-holes	through vs. blind holes	4
12	PHD	position and size of holes	bolt circle R30 vs. R42	77
13	CRS-A	angular spacing of features	equal-pitch vs. grouped holes	128
14	PAF	parallel planes, coplanarity	coplanar vs. stepped feet	31
15	MTS	wall thickness	1.5 mm vs. 3 mm wall	40
			Total	**781**

How the classical descriptors map onto this set. A3 and D3 are the angle and area readings of the same random triangle, so they form one block. Gaussian curvature (GCD), mean curvature (MCD), shape index (SID) and curvedness (CVD) are four views of the single joint (SI, CV) histogram (§5.2). The principal curvature ratio κ_2/κ_1 is a function of SI, so it is covered by the same block.

7.2 Assembling the final vector

1. **Histogram blocks** (D1, D2, triangle, D4, DAD, curvature, CRS-R, PHD images, MTS histogram): L1-normalise, then take the square root, $\hat{\mathbf{x}}_b = \sqrt{\mathbf{h}_b}$. Then $\|\hat{\mathbf{x}}_b\| = 1$ and

$$\langle \hat{\mathbf{x}}_b, \hat{\mathbf{x}}'_b \rangle = \sum_j \sqrt{h_j h'_j} = \text{Bhattacharyya similarity.}$$

2. **Other blocks** (SHPS, NCF, LBE, T1, CRS-A, PAF, and the scalar parts of DAD, D4 and MTS): standardise each value with the database mean and standard deviation, then scale the block to unit length.
3. **Weighted concatenation**:

$$\mathbf{x} = \frac{1}{\sqrt{\sum_b \beta_b}} \left[\sqrt{\beta_1}\, \hat{\mathbf{x}}_1,\ \ldots,\ \sqrt{\beta_{15}}\, \hat{\mathbf{x}}_{15} \right] \quad \Longrightarrow \quad \langle \mathbf{x}, \mathbf{x}' \rangle = \frac{\sum_b \beta_b \langle \hat{\mathbf{x}}_b, \hat{\mathbf{x}}'_b \rangle}{\sum_b \beta_b}. \tag{31}$$

   The overall similarity is then a weighted average of the fifteen block similarities. The weights β_b say how much each kind of information matters, and they can be tuned by leave-one-out retrieval accuracy on the labelled classes.
4. **Compression**: apply PCA per block (each block is small compared with the number of models), keep enough components per block to preserve retrieval accuracy, concatenate, and index with FAISS inner-product search.

7.3 Practical order of adoption

The ladder order explains *what* each block adds. For implementation, the most useful additions to an existing D1 + D2 + A3 pipeline are, in order: SHPS and DAD (exact, robust, and use information no point statistic has); CRS-R and CRS-A (turned and rotational parts); D3 and D4 (almost free once D2's samples exist); PAF and MTS (machined and moulded parts); T1 (a very strong categorical feature on watertight meshes); NCF; the curvature block; and finally LBE and PHD, which are the most powerful for connectivity and holes but also the most involved to compute.